EMPLOYEE DATA ANALYSIS DASHBOARD

Infanta Kennedy

ENTRI

21-DAY EXCEL DATA ANALYTICS CHALLENGE

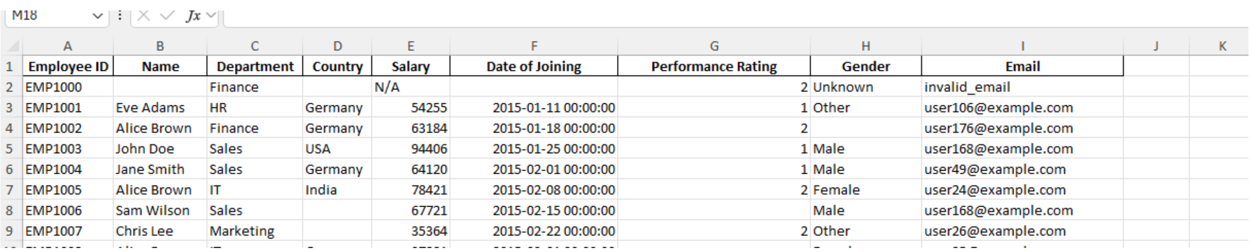
Day 1 – Dataset Selection & Import

Employee Details Dataset:

I have chosen Employee Details dataset here for reviewing and applying excel applications.

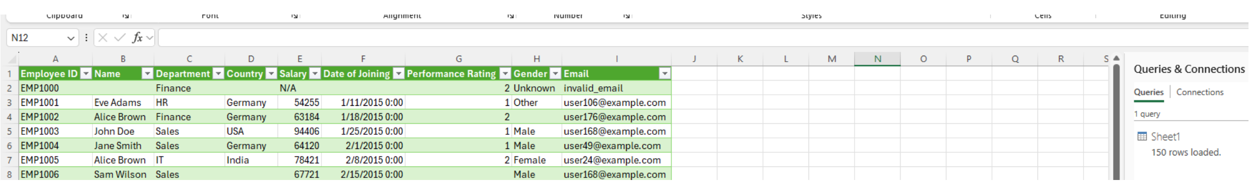
In order to import a excel dataset, I have first checked the type of excel file by clicking their property and load the data set by providing data->get data->from file->from excel_workbook

Dataset before Import:



Employee ID	Name	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1000		Finance		N/A		2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	54255	2015-01-11 00:00:00	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	63184	2015-01-18 00:00:00	2		user176@example.com
EMP1003	John Doe	Sales	USA	94406	2015-01-25 00:00:00	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	64120	2015-02-01 00:00:00	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	78421	2015-02-08 00:00:00	2	Female	user24@example.com
EMP1006	Sam Wilson	Sales		67721	2015-02-15 00:00:00		Male	user168@example.com
EMP1007	Chris Lee	Marketing		35364	2015-02-22 00:00:00	2	Other	user26@example.com

Dataset after Import:



Employee ID	Name	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1000		Finance		N/A		2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	54255	1/11/2015 0:00	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	63184	1/18/2015 0:00	2		user176@example.com
EMP1003	John Doe	Sales	USA	94406	1/25/2015 0:00	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	64120	2/1/2015 0:00	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	78421	2/8/2015 0:00	2	Female	user24@example.com
EMP1006	Sam Wilson	Sales		67721	2/15/2015 0:00		Male	user168@example.com

Key observations about the dataset:

1. The data set has employee details with several rows and columns.
2. It has missing data which is a blank data's
3. It has inconsistent data and duplicate data's
4. It has invalid data

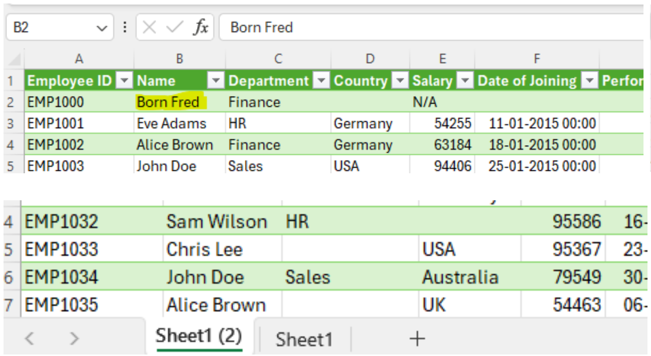
Day 2 – Data Entry & Formatting

Editing, inserting and deleting rows and columns

Editing:

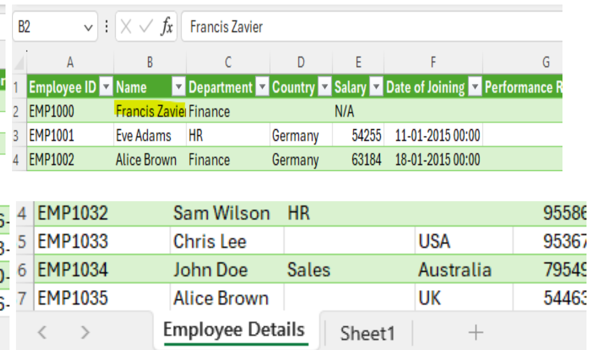
- We can edit the text by double clicking the cell and reenter/edit whatever details is required.

- We can also do it by selecting the current cell and entering the details in formula bar
- We can edit the sheets names as well like the above



	A	B	C	D	E	F
1	Employee ID	Name	Department	Country	Salary	Date of Joining
2	EMP1000	Born Fred	Finance		N/A	
3	EMP1001	Eve Adams	HR	Germany	54255	11-01-2015 00:00
4	EMP1002	Alice Brown	Finance	Germany	63184	18-01-2015 00:00
5	EMP1003	John Doe	Sales	USA	94406	25-01-2015 00:00

4	EMP1032	Sam Wilson	HR		95586	16-
5	EMP1033	Chris Lee		USA	95367	23-
6	EMP1034	John Doe	Sales	Australia	79549	30-
7	EMP1035	Alice Brown		UK	54463	06-

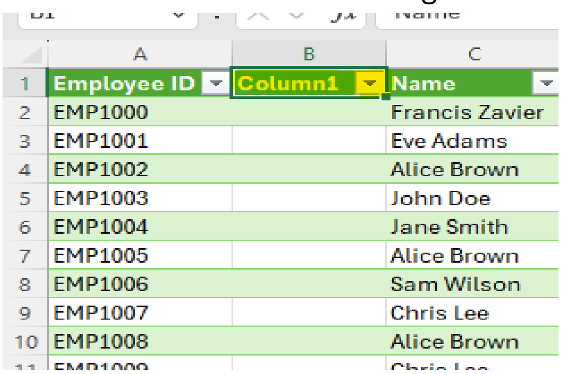


	A	B	C	D	E	F
1	Employee ID	Name	Department	Country	Salary	Date of Joining
2	EMP1000	Francis Zavier	Finance		N/A	
3	EMP1001	Eve Adams	HR	Germany	54255	11-01-2015 00:00
4	EMP1002	Alice Brown	Finance	Germany	63184	18-01-2015 00:00

4	EMP1032	Sam Wilson	HR		95586
5	EMP1033	Chris Lee		USA	95367
6	EMP1034	John Doe	Sales	Australia	79549
7	EMP1035	Alice Brown		UK	54463

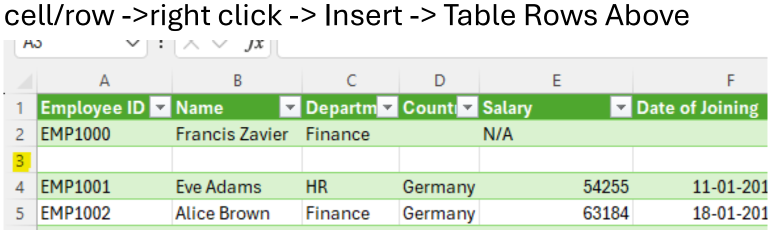
Inserting/Deleting Row/Column:

- To insert a column, select a cell/column -> Home -> Insert -> Insert columns to the left or select a cell/column ->right click -> Insert -> Table columns to the left



	A	B	C
1	Employee ID	Column1	Name
2	EMP1000		Francis Zavier
3	EMP1001		Eve Adams
4	EMP1002		Alice Brown
5	EMP1003		John Doe
6	EMP1004		Jane Smith
7	EMP1005		Alice Brown
8	EMP1006		Sam Wilson
9	EMP1007		Chris Lee
10	EMP1008		Alice Brown

- To insert a row, select a cell/row -> Home -> Insert -> Table Rows Above or select a cell/row ->right click -> Insert -> Table Rows Above



	A	B	C	D	E	F
1	Employee ID	Name	Departm	Count	Salary	Date of Joining
2	EMP1000	Francis Zavier	Finance		N/A	
3						
4	EMP1001	Eve Adams	HR	Germany	54255	11-01-201
5	EMP1002	Alice Brown	Finance	Germany	63184	18-01-201

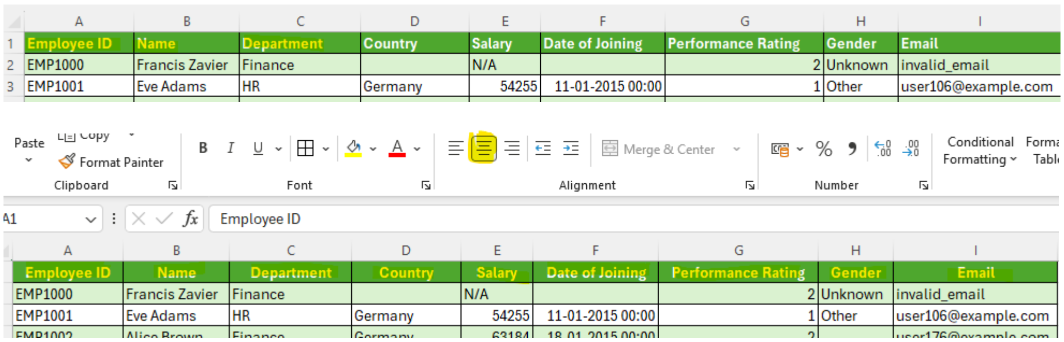
- To delete a column select a cell/column -> Home -> Delete-> Delete Sheet Columns or select a cell/column ->right click -> Delete -> Table columns
- To delete a row select a cell/row -> Home -> Delete-> Table Rows or select a cell/row ->right click -> Delete -> Delete Table Rows

Format headers (bold, colors, alignment)

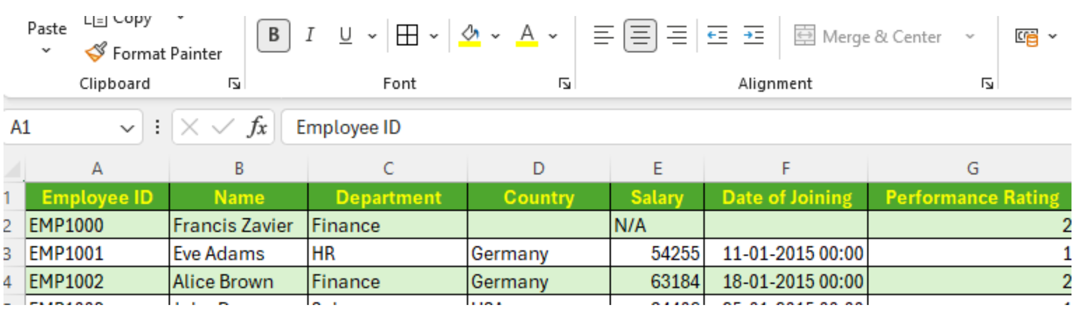
In the Home -> Font and Alignment tabs are used for aligning the datas like left, centre and right aligns, colouring the cells, making it bold, font style, font size, font style.

Examples:

Alignments:



Changing font Colour:



Day 3 – Basic Excel Formulas (Part 1)

Creating a summary table showing total, average, and count of main numeric columns:

I have chosen the **salary column** to perform the following functions.

1. Sum() : This function used to add all the salaries to provide total salary
2. Average() : This function used to find the average of the total salary found
3. Min() : This function used to find the minimum salary of the salary datas.
4. Max() : This function used to find the minimum salary of the salary datas.
5. Count() : This function used to count the number of salary entries which is in numeric data type.

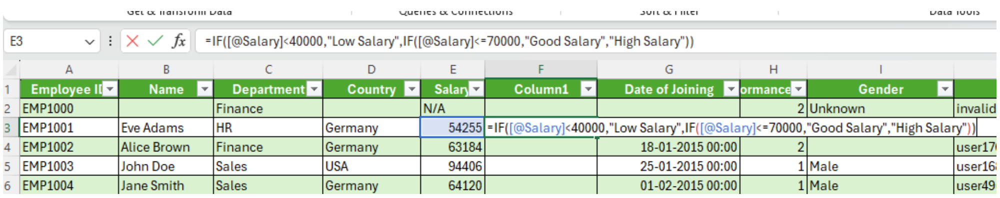
Summary

Sum	9411892
Average	66280.93
Min	30280
Max	99621
Count	142

Day 4 – Basic Excel Formulas (Part 2)

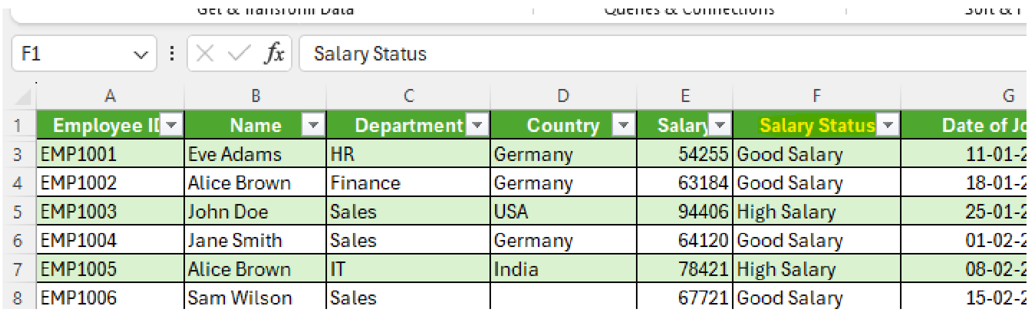
Writing at least 3 conditions

- **If()** : This condition function used to get results based on given criteria. Here, I have used nested if function to use multiple conditions in a single formula for a salary column to find the salary status whether it falls under low, good or high salary.



E3 =IF([@Salary]<40000,"Low Salary",IF([@Salary]<=70000,"Good Salary","High Salary"))

	Employee II	Name	Department	Country	Salary	Column1	Date of Joining	ormance	Gender	
2	EMP1000		Finance		N/A			2	Unknown	invalid
3	EMP1001	Eve Adams	HR	Germany	54255	=IF([@Salary]<40000,"Low Salary",IF([@Salary]<=70000,"Good Salary","High Salary"))				
4	EMP1002	Alice Brown	Finance	Germany	63184		18-01-2015 00:00	2		user17
5	EMP1003	John Doe	Sales	USA	94406		25-01-2015 00:00	1	Male	user16
6	EMP1004	Jane Smith	Sales	Germany	64120		01-02-2015 00:00	1	Male	user49



F1 Salary Status

	Employee II	Name	Department	Country	Salary	Salary Status	Date of Jo
3	EMP1001	Eve Adams	HR	Germany	54255	Good Salary	11-01-2
4	EMP1002	Alice Brown	Finance	Germany	63184	Good Salary	18-01-2
5	EMP1003	John Doe	Sales	USA	94406	High Salary	25-01-2
6	EMP1004	Jane Smith	Sales	Germany	64120	Good Salary	01-02-2
7	EMP1005	Alice Brown	IT	India	78421	High Salary	08-02-2
8	EMP1006	Sam Wilson	Sales		67721	Good Salary	15-02-2

- Sumif() : This function used to sum values based on specific condition
- Countif() : This function used to count the occurances based on given condition
- Averageif() : This function used to provide average of values based on condition

Total HR Salary	1630404
Germany Count	23
Average HR Salary	67834.57

Day 5 – Text Functions Practice

Cleaning and formatting the text columns (names, cities, etc.) properly.

- LEFT() – This function used to truncate or limit the number of characters displayed from a left side of a text string. =LEFT(text, [number of chars])
- RIGHT() - This function used to truncate or limit the number of characters displayed from a right side of a text string. =RIGHT(text, [number of chars])
- MID() – Used to extract a substring from a middle of a text string. =MID(strat num, num chars)
- UPPER() – Coverts text to a upper case
- LOWER() – Converts text to a lower case
- PROPER() – Converts text with first letter capitalized

In my dataset, I don't have any columns to do the above functions, therefore skipping it.

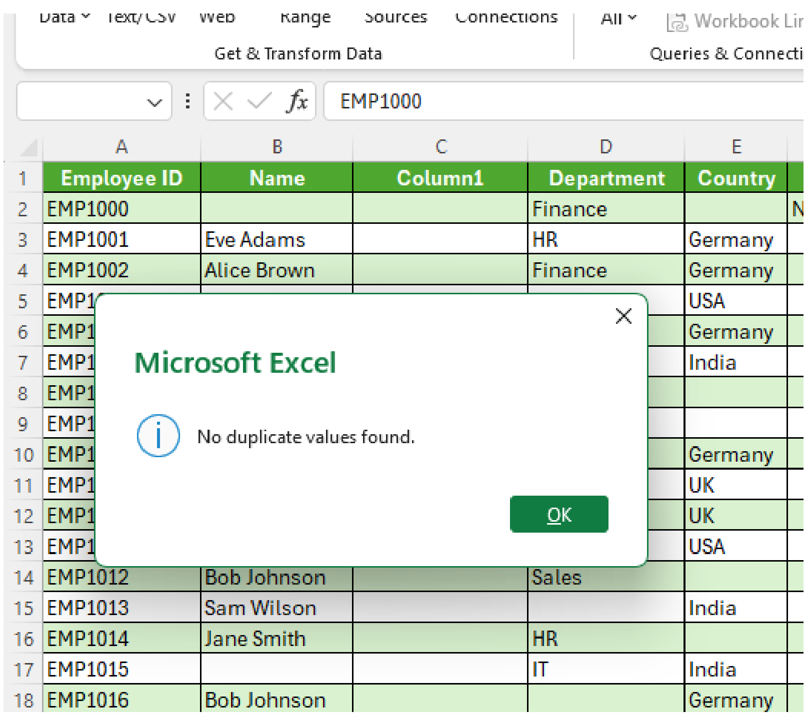
Day 6 – Data Validation & Cleaning Intro

Ensure your dataset has no duplicates and missing values handled

Firstly, I am going to check the duplicates, for this task, I selected the entire table -> In Data -> Select -> Remove Duplicates -> Select all the columns

In this case, I couldn't find the duplicates

Now, I am trying another option to check and remove duplicates by considering the Employee id as a unique value. But, the result was no duplicates found. So, I am keeping the file as it is.



Firstly, I am going to find the blank values by checking the filter option in column headers one by one.

In Name column – I deleted the rows which has blank names as its not adding value to my analysis as those blank ones, doesn't contain useful information regarding the other columns as well

Employee ID	Name	Department	Country	Salary	Salary Status	Date of Joining
EMP1000		Finance		N/A	High Salary	
EMP1015		IT	India	92443	High Salary	19-04-2015 00:00
EMP1030		Marketing		92259	High Salary	
EMP1045		HR		58962	Good Salary	15-11-2015 00:00
EMP1060		Marketing	Germany	N/A	High Salary	
EMP1075		Sales		96142	High Salary	12-06-2016 00:00
EMP1090		Marketing	UK	50741	Good Salary	25-09-2016 00:00
EMP1105		HR	UK	87963	High Salary	08-01-2017 00:00
EMP1120		IT	USA	N/A	High Salary	
EMP1135		IT	UK	39280	Low Salary	06-08-2017 00:00

Get & Transform Data   Queries & Connections   Sort & Filt

A2   EMP1001

	Employee ID	Name	Department	Country	Salary	Salary Status	Date of J
2							
3							
4							
5							
6							

In Department and Country column – I found a blank values and replaced with their mode(most frequent value) as it belongs to categorical values

C14   Marketing

	Employee ID	Name	Department	Country	Salary	
14	EMP1013	Sam Wilson	Marketing	India	62158	Goc
16	EMP1016	Bob Johnson	Marketing	Germany	90271	Higl
19	EMP1019	Eve Adams	Marketing	India	67531	Goc
25	EMP1025	Chris Lee	Marketing	India	52272	Goc
32	EMP1033	Chris Lee	Marketing	USA	95367	Higl
34	EMP1035	Alice Brown	Marketing	UK	54463	Goc
41	EMP1042	Bob Johnson	Marketing	Australia	64913	Goc
48	EMP1050	Chris Lee	Marketing	India	30686	Low
53	EMP1055	Chris Lee	Marketing	USA	74815	Higl

D7   India

1	Employee ID	Name	Department	Country	Salary	Salary Status
7	EMP1006	Sam Wilson	Sales	India	67721	Good Salary
8	EMP1007	Chris Lee	Marketing	India	35364	Low Salary
13	EMP1012	Bob Johnson	Sales	India	96418	High Salary
15	EMP1014	Jane Smith	HR	India	59894	Good Salary
21	EMP1021	Jane Smith	HR	India	98922	High Salary
24	EMP1024	Sam Wilson	Finance	India	91005	High Salary
25	EMP1025	Chris Lee	Unknown	India	52272	Good Salary
31	EMP1032	Sam Wilson	HR	India	95586	High Salary
36	EMP1037	Alice Brown	HR	India	64796	Good Salary
43	EMP1044	Chris Lee	Finance	India	81610	High Salary
46	EMP1048	John Doe	Finance	India	66513	Good Salary
48	EMP1050	Chris Lee	Unknown	India	30686	Low Salary

In salary Column – I found the blanks and replaced it with imputation average of salary which is 65882.24

	A	B	C	D	E	F
	Employee ID	Name	Department	Country	Salary	Salary Status
	:MP1020	Jane Smith	HR	USA	65882	Good Salary
	:MP1040	Bob Johnson	Finance	UK	65882	Good Salary
	:MP1080	Eve Adams	Marketing	India	65882	Good Salary
	:MP1100	Bob Johnson	Marketing	India	65882	Good Salary
	:MP1140	Chris Lee	Sales	USA	65882	Good Salary

In the date of joining column, it has 5 blanks which is very few values are missing. When I reviewed the other employee date of joining detail column it has a series of 7 days added to it, therefore I am imputing the DOJ with those values.

Example:

The row 9 has a blank DOJ, but you see the previous entries it has as series of 7 days added to the preceding dates, so I am filling it as (previous date is 4 and added 7 to get date 11, which in addition if we add 7 to 11 you will get 18. I followed the same for all 5 blank values

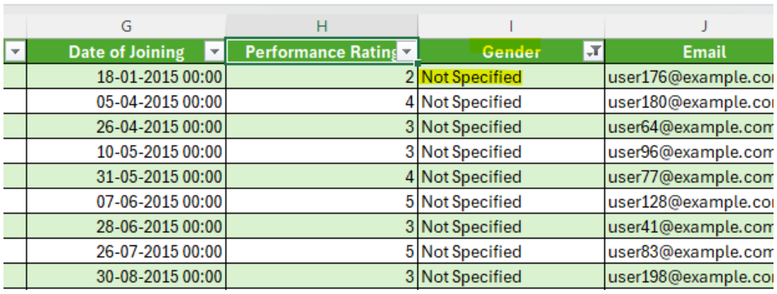
3	EMP1034	John Doe	Sales	Australia	79549	High Salary	30-08-2015 00:00	3	
4	EMP1035	Alice Brown	Marketing	UK	54463	Good Salary	06-09-2015 00:00	4	Male
5	EMP1036	Alice Brown	IT	USA	89781	High Salary	13-09-2015 00:00		Female
6	EMP1037	Alice Brown	HR	India	64796	Good Salary	20-09-2015 00:00		Other
7	EMP1038	John Doe	Marketing	UK	52093	Good Salary	27-09-2015 00:00	5	
8	EMP1039	Chris Lee	HR	UK	48224	Good Salary	04-10-2015 00:00	3	
9	EMP1040	Bob Johnson	Finance	UK	65882	Good Salary		4	Female
0	EMP1041	Alice Brown	IT	Germany	33956	Low Salary	18-10-2015 00:00		
1	EMP1042	Bob Johnson	Marketing	Australia	64913	Good Salary	25-10-2015 00:00	1	Female
2	EMP1043	Alice Brown	Marketing	Australia	35326	Low Salary	01-11-2015 00:00		Female
3	EMP1044	Chris Lee	Finance	India	81610	High Salary	08-11-2015 00:00	1	
4	EMP1046	Chris Lee	IT	Germany	92701	High Salary	22-11-2015 00:00	3	Other
5	EMP1047	Chris Lee	HR	India	58245	Good Salary	29-11-2015 00:00	4	

5	EMP1036	Alice Brown	IT	USA	89781	High Salary	13-09-2015 00:00
6	EMP1037	Alice Brown	HR	India	64796	Good Salary	20-09-2015 00:00
7	EMP1038	John Doe	Marketing	UK	52093	Good Salary	27-09-2015 00:00
8	EMP1039	Chris Lee	HR	UK	48224	Good Salary	04-10-2015 00:00
9	EMP1040	Bob Johnson	Finance	UK	65882	Good Salary	11-10-2015 00:00
0	EMP1041	Alice Brown	IT	Germany	33956	Low Salary	18-10-2015 00:00
1	EMP1042	Bob Johnson	Marketing	Australia	64913	Good Salary	25-10-2015 00:00
2	EMP1043	Alice Brown	Marketing	Australia	35326	Low Salary	01-11-2015 00:00
3	EMP1044	Chris Lee	Finance	India	81610	High Salary	08-11-2015 00:00

In performance rating column – I used the median value to fill the blank values which is 3 here in a rating of (1-5)

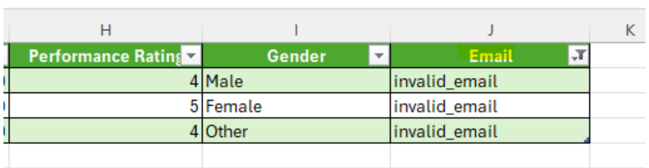
	F	G	H	I	J
	Salary Status	Date of Joining	Performance Rating	Gender	Emai
57721	Good Salary	15-02-2015 00:00	3	Male	user168@e
97881	High Salary	01-03-2015 00:00	3	Female	user25@ex
91846	High Salary	08-03-2015 00:00	3	Male	user153@e
90271	High Salary	26-04-2015 00:00	3		user64@ex
73602	High Salary	10-05-2015 00:00	3		user96@ex
95367	High Salary	23-08-2015 00:00	3	Male	user136@e
39781	High Salary	13-09-2015 00:00	3	Female	user31@ex
54796	Good Salary	20-09-2015 00:00	3	Other	user98@ex

In Gender column – we have so many blanks and we couldn't use the mode imputation as well, as the other categories falls are same values, so for safer side, I am filling it as not-specified.

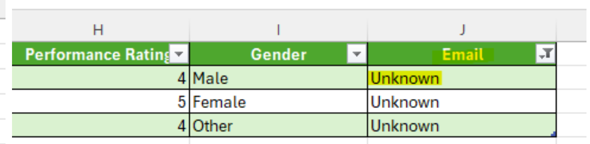


Date of Joining	Performance Rating	Gender	Email
18-01-2015 00:00	2	Not Specified	user176@example.co
05-04-2015 00:00	4	Not Specified	user180@example.co
26-04-2015 00:00	3	Not Specified	user64@example.com
10-05-2015 00:00	3	Not Specified	user96@example.com
31-05-2015 00:00	4	Not Specified	user77@example.com
07-06-2015 00:00	5	Not Specified	user128@example.co
28-06-2015 00:00	3	Not Specified	user41@example.com
26-07-2015 00:00	5	Not Specified	user83@example.com
30-08-2015 00:00	3	Not Specified	user198@example.co

In Email id column – I have identified 3 values which has written as invalid_email, therefore I am filling it as Unknown for safer side as email id is unique value and I couldn't find any other details/patterns to fill it.



Performance Rating	Gender	Email
4	Male	invalid_email
5	Female	invalid_email
4	Other	invalid_email



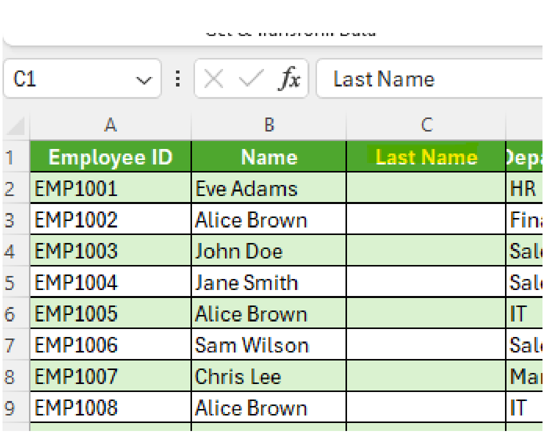
Performance Rating	Gender	Email
4	Male	Unknown
5	Female	Unknown
4	Other	Unknown

WEEK 2 – Data Cleaning, Exploration & Analysis

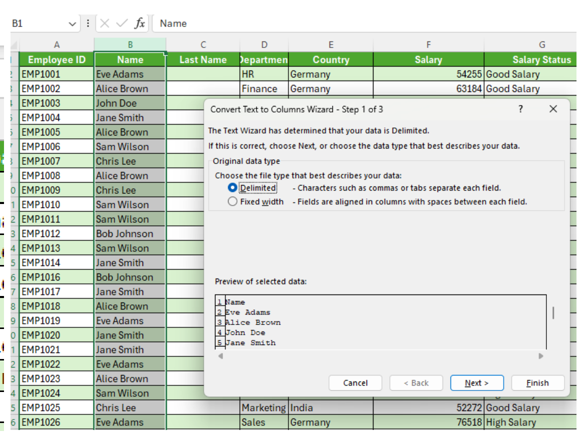
Day 8 – Handling Inconsistent Data

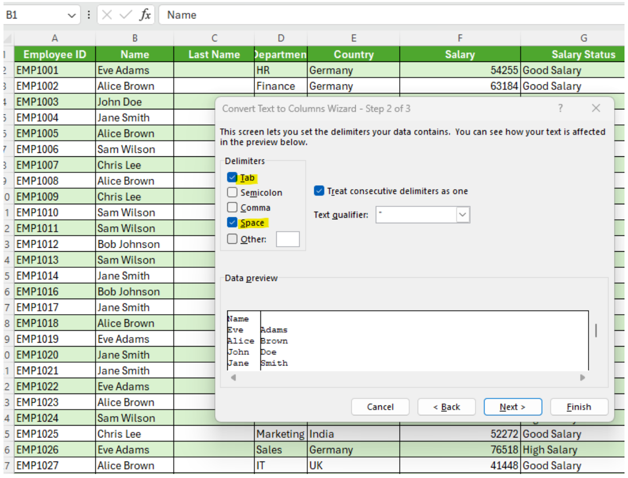
- **Split one messy column into multiple clean ones**

I have split the Name column to First and Last name using **text to columns** function. Create new column ->Select the name column -> In Data tab -> Select text to columns -> Delimited -> Click Next -> Check in tab and space -> Next -> Convert the columns datatype to text ->Ok

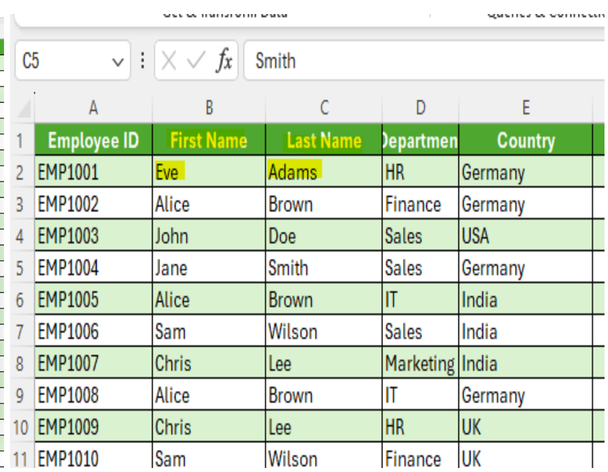


	Employee ID	Name	Last Name
2	EMP1001	Eve Adams	
3	EMP1002	Alice Brown	
4	EMP1003	John Doe	
5	EMP1004	Jane Smith	
6	EMP1005	Alice Brown	
7	EMP1006	Sam Wilson	
8	EMP1007	Chris Lee	
9	EMP1008	Alice Brown	



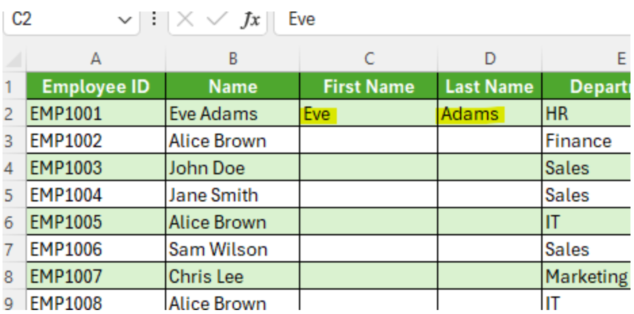


Employee ID	Name	Last Name	Department	Country	Salary	Salary Status
EMP1001	Eve Adams		HR	Germany	54255	Good Salary
EMP1002	Alice Brown		Finance	Germany	63184	Good Salary
EMP1003	John Doe					
EMP1004	Jane Smith					
EMP1005	Alice Brown					
EMP1006	Sam Wilson					
EMP1007	Chris Lee					
EMP1008	Alice Brown					
EMP1009	Chris Lee					
EMP1010	Sam Wilson					
EMP1011	Sam Wilson					
EMP1012	Bob Johnson					
EMP1013	Sam Wilson					
EMP1014	Jane Smith					
EMP1016	Bob Johnson					
EMP1017	Jane Smith					
EMP1018	Alice Brown					
EMP1019	Eve Adams					
EMP1020	Jane Smith					
EMP1021	Jane Smith					
EMP1022	Eve Adams					
EMP1023	Alice Brown					
EMP1024	Sam Wilson					
EMP1025	Chris Lee		Marketing	India	52272	Good Salary
EMP1026	Eve Adams		Sales	Germany	76518	High Salary
EMP1027	Alice Brown		IT	UK	41448	Good Salary



Employee ID	First Name	Last Name	Department	Country
EMP1001	Eve	Adams	HR	Germany
EMP1002	Alice	Brown	Finance	Germany
EMP1003	John	Doe	Sales	USA
EMP1004	Jane	Smith	Sales	Germany
EMP1005	Alice	Brown	IT	India
EMP1006	Sam	Wilson	Sales	India
EMP1007	Chris	Lee	Marketing	India
EMP1008	Alice	Brown	IT	Germany
EMP1009	Chris	Lee	HR	UK
EMP1010	Sam	Wilson	Finance	UK

We can also use Flash Fill to split names column to first and last name, by providing inputs to the first row and click CTRL + E. The excel with analyze the patterns and fill in the data.



Employee ID	Name	First Name	Last Name	Department
EMP1001	Eve Adams	Eve	Adams	HR
EMP1002	Alice Brown			Finance
EMP1003	John Doe			Sales
EMP1004	Jane Smith			Sales
EMP1005	Alice Brown			IT
EMP1006	Sam Wilson			Sales
EMP1007	Chris Lee			Marketing
EMP1008	Alice Brown			IT

Employee ID	Name	First Name	Last Name	Department
EMP1001	Eve Adams	Eve	Adams	HR
EMP1002	Alice Brown	Alice	Brown	Finance
EMP1003	John Doe	John	Doe	Sales
EMP1004	Jane Smith	Jane	Smith	Sales
EMP1005	Alice Brown	Alice	Brown	IT
EMP1006	Sam Wilson	Sam	Wilson	Sales
EMP1007	Chris Lee	Chris	Lee	Marketing
EMP1008	Alice Brown	Alice	Brown	IT

Trim() function used to remove leading spaces, removes trailing spaces, removes extra spaces between words, keeps only a single space between words.

My dataset doesn't have columns/datas to use TRIM() function

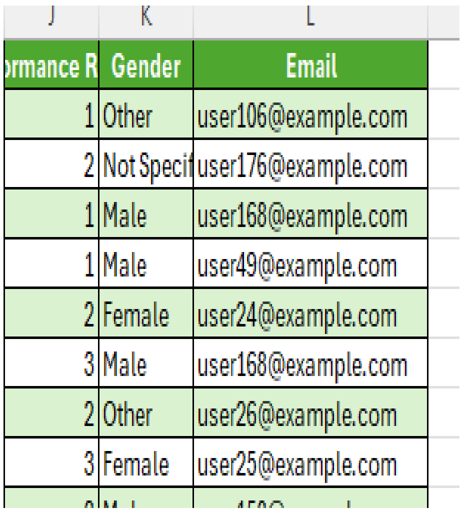
Day 9 – Conditional Formatting for Data Cleaning

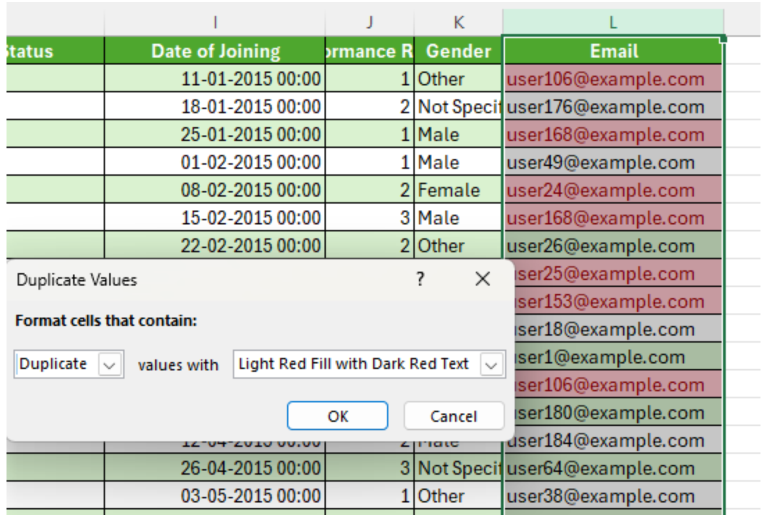
Highlight missing, duplicate, or outlier values by creating color-based rules to visually flag data issues.

In order to find the duplicates, please find below steps:

Select the column/table where we need to find duplicates

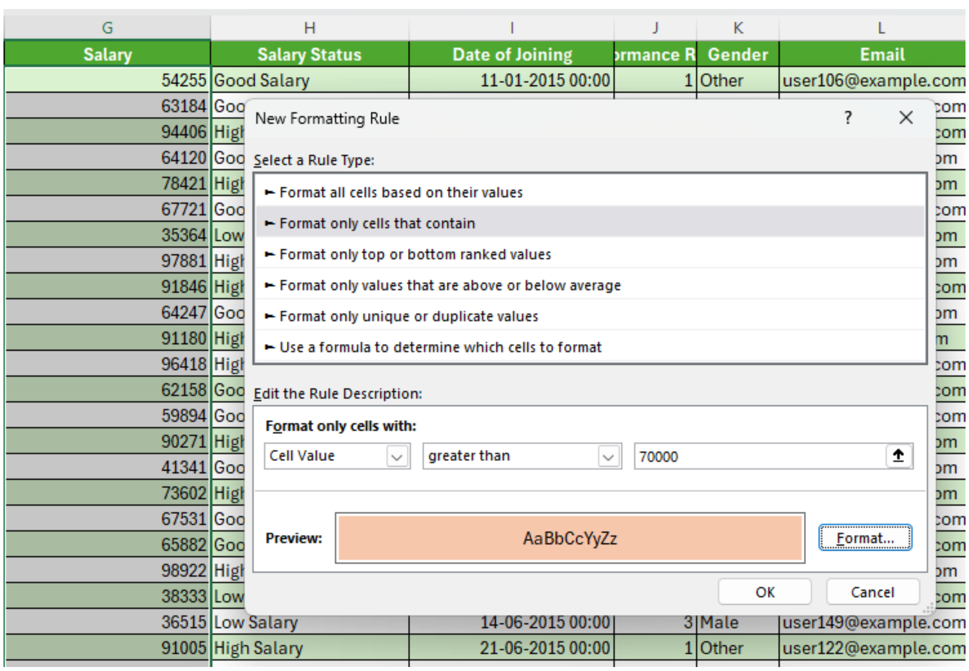
In Home -> Styles tab -> Select Conditional Formatting -> Highlight Cells Rules -> Duplicate Values -> Choose the color you want the duplicates to be highlighted -> Click Ok





If we want to set rules based on some conditions also, we can use the above steps andthen to choose the more rules function.

In below example, we can find salary greater than 70,000 to be highlighted



Result:

Salary	Salary Status	Date of Joining	rmanc
54255	Good Salary	11-01-2015 00:00	
63184	Good Salary	18-01-2015 00:00	
94406	High Salary	25-01-2015 00:00	
64120	Good Salary	01-02-2015 00:00	
78421	High Salary	08-02-2015 00:00	
67721	Good Salary	15-02-2015 00:00	
35364	Low Salary	22-02-2015 00:00	
97881	High Salary	01-03-2015 00:00	
91846	High Salary	08-03-2015 00:00	
64247	Good Salary	15-03-2015 00:00	
91180	High Salary	22-03-2015 00:00	
96418	High Salary	29-03-2015 00:00	
62158	Good Salary	05-04-2015 00:00	
59894	Good Salary	12-04-2015 00:00	
90271	High Salary	26-04-2015 00:00	
41341	Good Salary	03-05-2015 00:00	
73602	High Salary	10-05-2015 00:00	
67531	Good Salary	17-05-2015 00:00	
65882	Good Salary	24-05-2015 00:00	

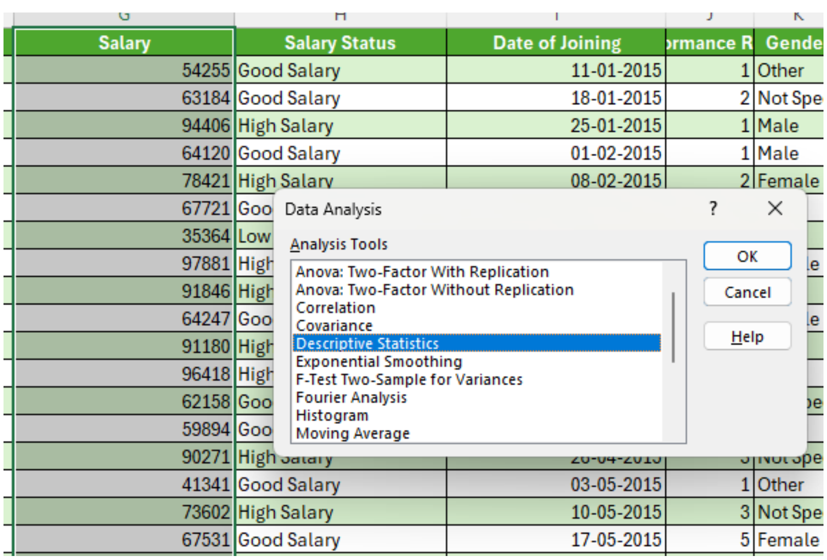
Day 10 – Descriptive Statistics

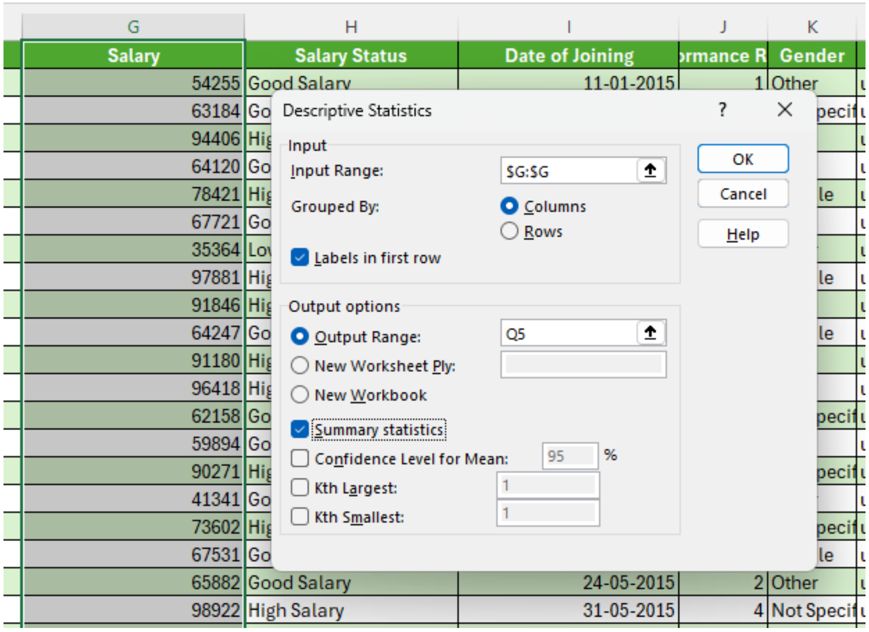
● Use Analysis ToolPak → Descriptive Statistics.

Task: Generate Mean, Median, and Standard Deviation for numeric columns.

I have only salary column as a numeric value, therefore I am going to fin dthe descriptive statics for the salary column.

I have added the data analysis tab by adding add-ins -> ToolPak





Summary;

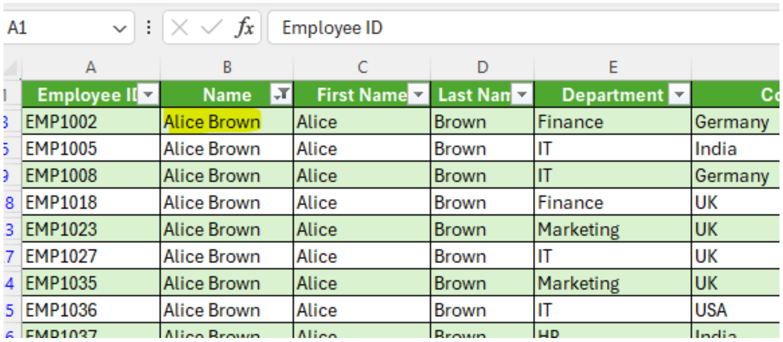
Salary	
Mean	65882.22857
Standard Error	1788.068907
Median	65980.5
Mode	65882
Standard Deviation	21156.71662
Sample Variance	447606658.2
Kurtosis	-1.199401084
Skewness	-0.108481194
Range	69341
Minimum	30280
Maximum	99621
Sum	9223512
Count	140

Day 11 – Filtering, Sorting & Exploring Data

- Apply multiple filters and sorts.

In Data ribbon -> select Filter to apply filters in every column. We can easily filter the data by clicking the header filter arrow and by selecting the required data.

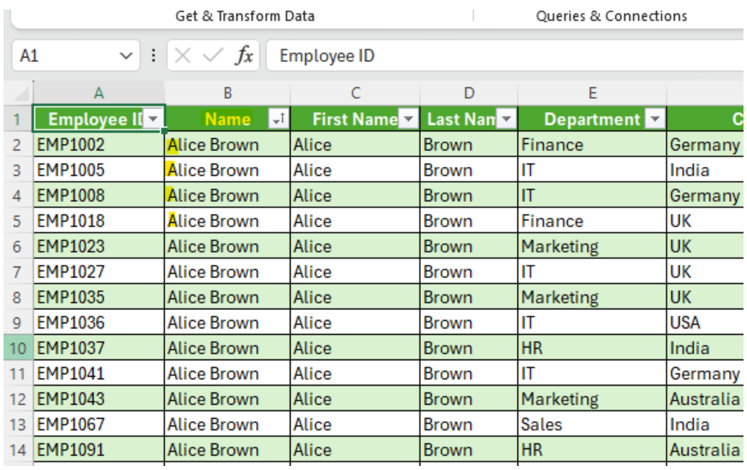
Example, I have filtered the name Alice Brown and all the row/column data which has name Alice Brown are retrieved



A1 | Employee ID

Employee ID	Name	First Name	Last Name	Department	Co
EMP1002	Alice Brown	Alice	Brown	Finance	Germany
EMP1005	Alice Brown	Alice	Brown	IT	India
EMP1008	Alice Brown	Alice	Brown	IT	Germany
EMP1018	Alice Brown	Alice	Brown	Finance	UK
EMP1023	Alice Brown	Alice	Brown	Marketing	UK
EMP1027	Alice Brown	Alice	Brown	IT	UK
EMP1035	Alice Brown	Alice	Brown	Marketing	UK
EMP1036	Alice Brown	Alice	Brown	IT	USA

Also, if we want to sort any columns, we can choose Sort (A-Z) for sorting in ascending order and Sort(Z-A) in descending order



Get & Transform Data | Queries & Connections

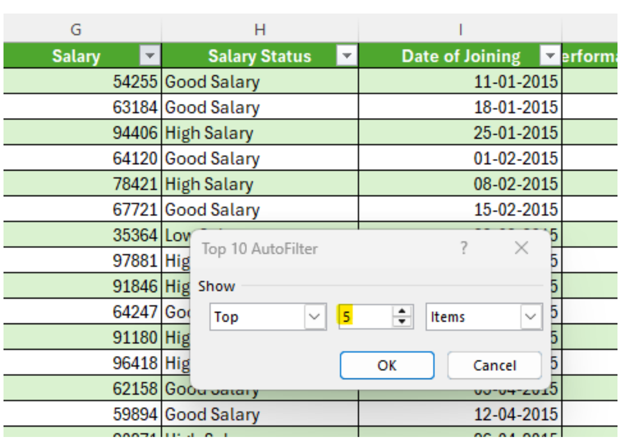
A1 | Employee ID

	Employee ID	Name	First Name	Last Name	Department	C
2	EMP1002	Alice Brown	Alice	Brown	Finance	Germany
3	EMP1005	Alice Brown	Alice	Brown	IT	India
4	EMP1008	Alice Brown	Alice	Brown	IT	Germany
5	EMP1018	Alice Brown	Alice	Brown	Finance	UK
6	EMP1023	Alice Brown	Alice	Brown	Marketing	UK
7	EMP1027	Alice Brown	Alice	Brown	IT	UK
8	EMP1035	Alice Brown	Alice	Brown	Marketing	UK
9	EMP1036	Alice Brown	Alice	Brown	IT	USA
10	EMP1037	Alice Brown	Alice	Brown	HR	India
11	EMP1041	Alice Brown	Alice	Brown	IT	Germany
12	EMP1043	Alice Brown	Alice	Brown	Marketing	Australia
13	EMP1067	Alice Brown	Alice	Brown	Sales	India
14	EMP1091	Alice Brown	Alice	Brown	HR	Australia

- Find top 5 and bottom 5 entries by value. Task: Present a mini report of key patterns (Top 5 salary, etc.)

I have chosen the salary column, to find top 5 salary list

Give filter to the columns -> In salary column choose number filters -> Top 10, where we can edit to top 5



Salary	Salary Status	Date of Joining
54255	Good Salary	11-01-2015
63184	Good Salary	18-01-2015
94406	High Salary	25-01-2015
64120	Good Salary	01-02-2015
78421	High Salary	08-02-2015
67721	Good Salary	15-02-2015
35364	Low	
97881	Hig	
91846	Hig	
64247	Goo	
91180	Hig	
96418	Hig	
62158	Good Salary	
59894	Good Salary	12-04-2015

Result : Top 5 Salaries

Employee II	Name	First Name	Last Nan	Department	Country	Salary	Salary Status	Date of Joining	erformance Ra	Gender	Email
EMP1008	Alice Brown	Alice	Brown	IT	Germany	97881	High Salary	01-03-2015	3	Female	user25@example.com
EMP1021	Jane Smith	Jane	Smith	HR	India	98922	High Salary	31-05-2015	4	Not Specified	user77@example.com
EMP1062	Chris Lee	Chris	Lee	IT	UK	97981	High Salary	13-03-2016	5	Female	user71@example.com
EMP1083	Sam Wilson	Sam	Wilson	Sales	India	99621	High Salary	07-08-2016	3	Other	user145@example.com
EMP1149	Sam Wilson	Sam	Wilson	Marketing	India	98475	High Salary	12-11-2017	5	Other	user30@example.com

Day 12 – Date & Time Functions

Practice: TODAY(), NOW(), YEAR(), MONTH(), WEEKDAY(), DATEDIF()

Task: Calculate age, duration, or time differences from your dataset.

TODAY() – used to return the today's date

NOW() – used to return the date along with time

YEAR() – used to return year of the selected date

MONTH() – used to find the month in a date

WEEKDAY() – used to find the week day of the given date

	Date functions	
	Today()	17-05-2026
	Now()	17-05-2026 00:19
Year	11-01-2015	2015
Month	11-01-2015	1
Weekday	11-01-2015	1

DATEDIF() – used to find the difference between two dates/months/years

Date of Joining	No. of Days with Employer	formance Rat	
11-01-2015	=datedif([@[Date of Joining]],TODAY(),"d")		Oth
18-01-2015		2	Not
25-01-2015		1	Mal
01-02-2015		1	Mal

Result:

Date of Joining	No. of Days with Employer	form
11-01-2015	4144	
18-01-2015	4137	
25-01-2015	4130	
01-02-2015	4123	
08-02-2015	4116	

Day 13 – Lookup & Reference Functions (Part 1)

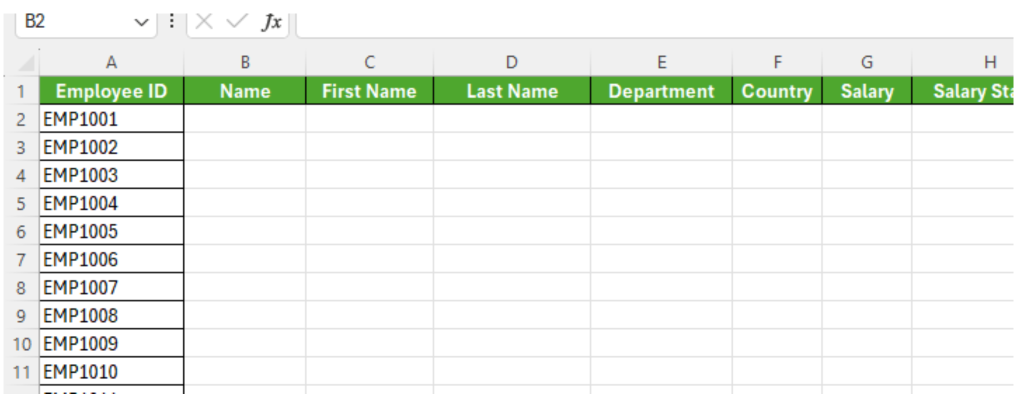
Use: VLOOKUP() Task: Duplicate the dataset using Vlookup function

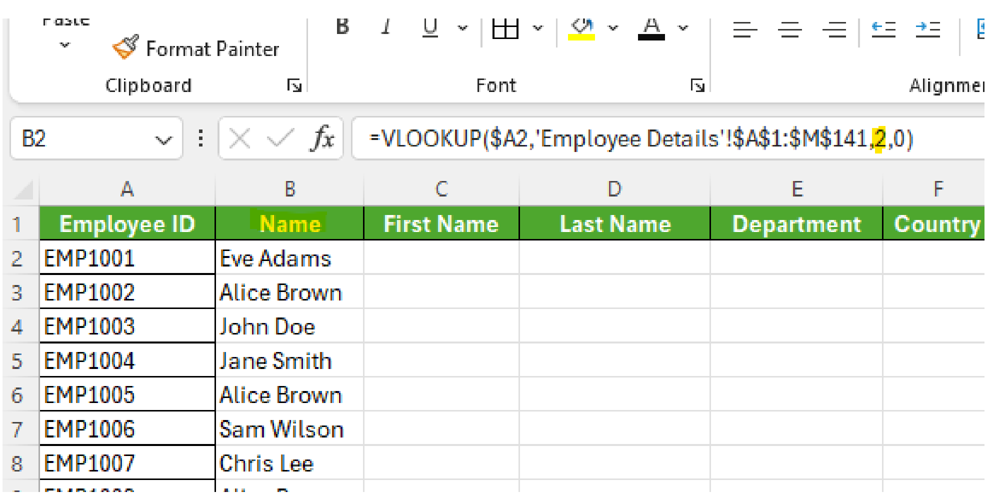
Vlookup() is powerful function in excel that help you search for specific value in a table and retrieve information based on that value.

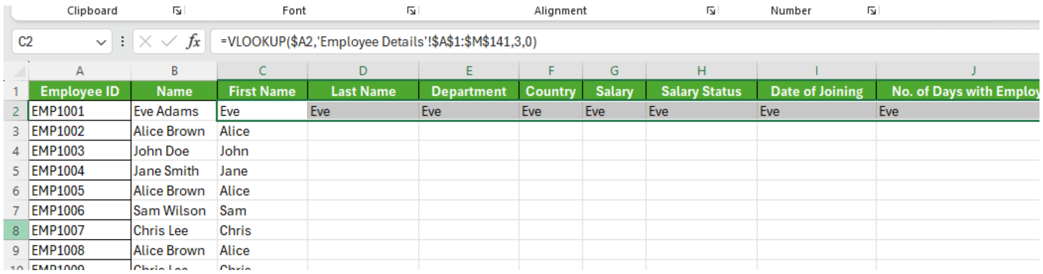
Syntax: VLOOKUP(lookup_value, table_array, col_index_num,[range_lookup])

In order to perform the above task, duplicating the dataset, I added a new sheet named Vlookup for creating a duplicate one.

1. Copy pasted the Emp id column as a unique value
2. I have used the vlookup formula for the second column “Name” by providing the column index number as 2 and used absolute reference “$” to use the same formula for veery columns by just dragging it and changing the corresponding column index values as 3, 4, 5,6 ,7,etc.

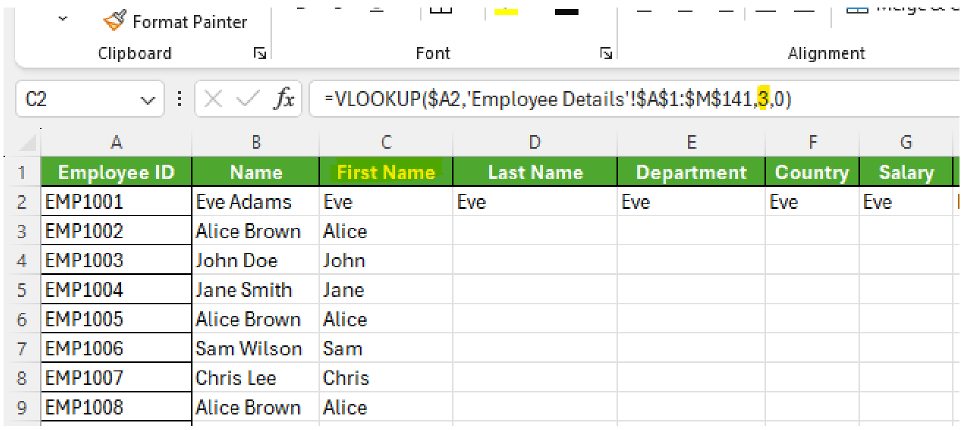






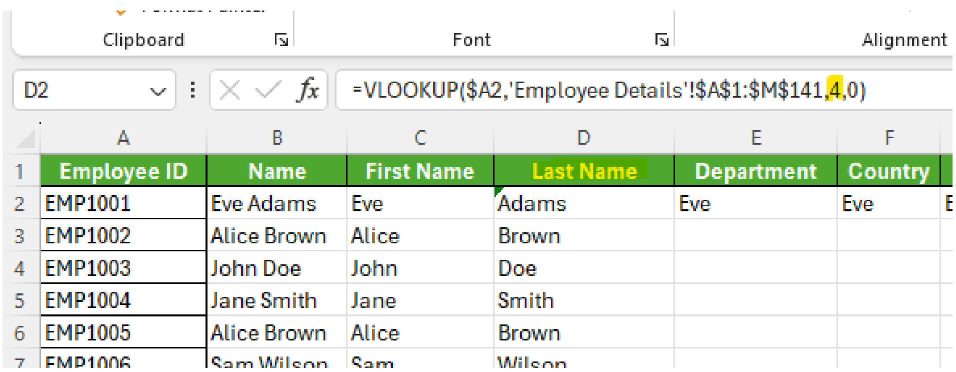
C2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,3,0)

Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining	No. of Days with Employ
EMP1001	Eve Adams	Eve	Eve	Eve	Eve	Eve	Eve	Eve	Eve
EMP1002	Alice Brown	Alice							
EMP1003	John Doe	John							
EMP1004	Jane Smith	Jane							
EMP1005	Alice Brown	Alice							
EMP1006	Sam Wilson	Sam							
EMP1007	Chris Lee	Chris							
EMP1008	Alice Brown	Alice							



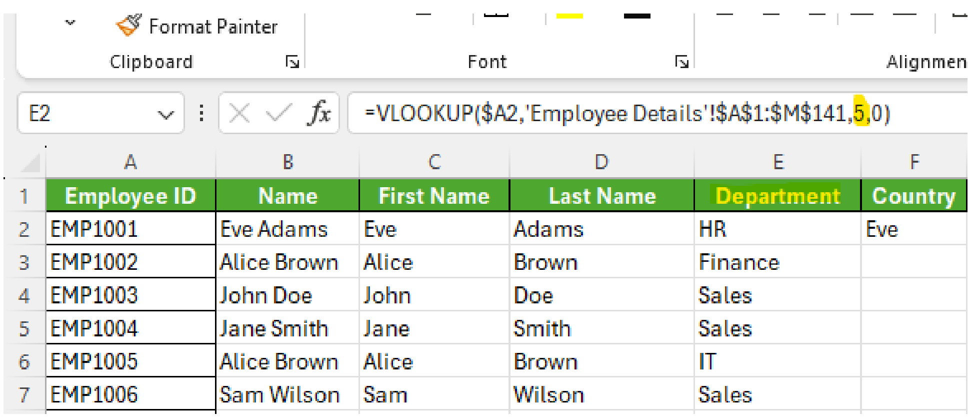
C2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,3,0)

Employee ID	Name	First Name	Last Name	Department	Country	Salary
EMP1001	Eve Adams	Eve	Eve	Eve	Eve	Eve
EMP1002	Alice Brown	Alice				
EMP1003	John Doe	John				
EMP1004	Jane Smith	Jane				
EMP1005	Alice Brown	Alice				
EMP1006	Sam Wilson	Sam				
EMP1007	Chris Lee	Chris				
EMP1008	Alice Brown	Alice				



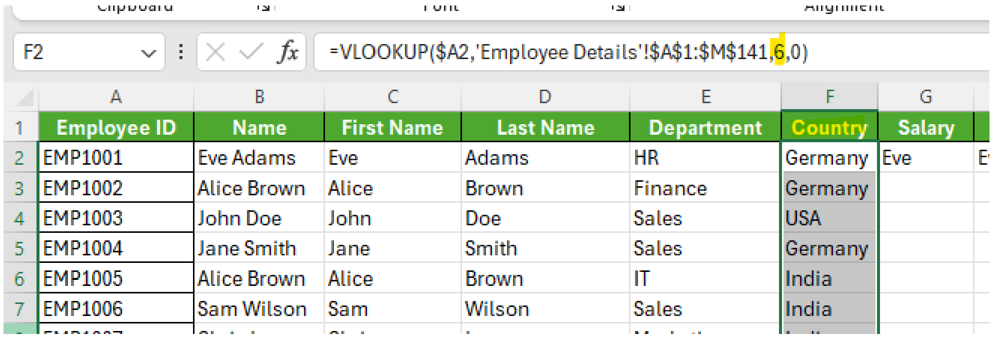
D2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,4,0)

Employee ID	Name	First Name	Last Name	Department	Country
EMP1001	Eve Adams	Eve	Adams	Eve	Eve
EMP1002	Alice Brown	Alice	Brown		
EMP1003	John Doe	John	Doe		
EMP1004	Jane Smith	Jane	Smith		
EMP1005	Alice Brown	Alice	Brown		



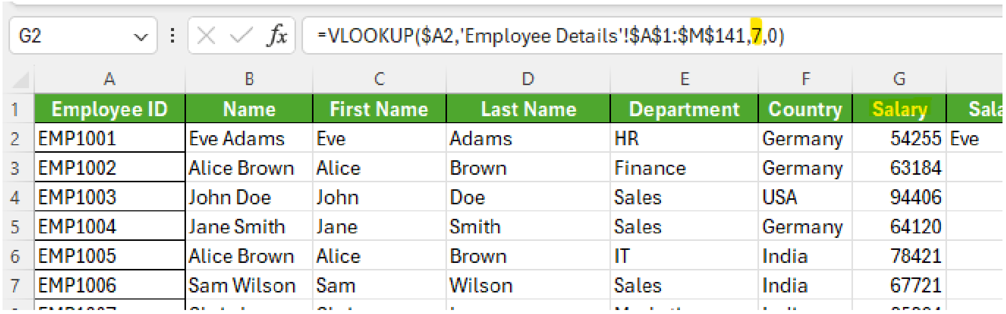
E2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,5,0)

Employee ID	Name	First Name	Last Name	Department	Country
EMP1001	Eve Adams	Eve	Adams	HR	Eve
EMP1002	Alice Brown	Alice	Brown	Finance	
EMP1003	John Doe	John	Doe	Sales	
EMP1004	Jane Smith	Jane	Smith	Sales	
EMP1005	Alice Brown	Alice	Brown	IT	
EMP1006	Sam Wilson	Sam	Wilson	Sales	



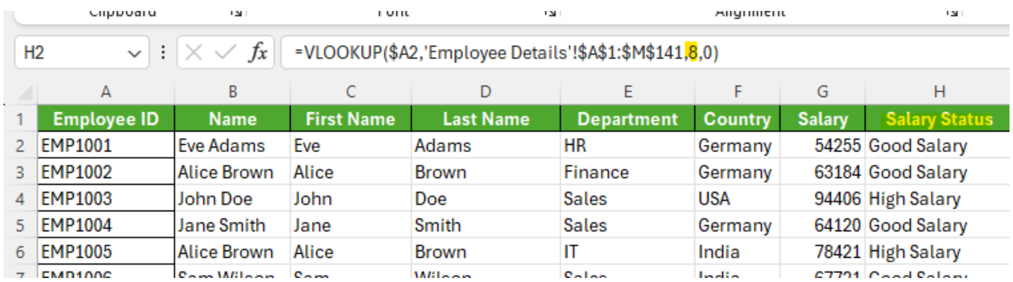
F2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,6,0)

	A	B	C	D	E	F	G
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	Eve
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	
4	EMP1003	John Doe	John	Doe	Sales	USA	
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	
6	EMP1005	Alice Brown	Alice	Brown	IT	India	
7	EMP1006	Sam Wilson	Sam	Wilson	Sales	India	



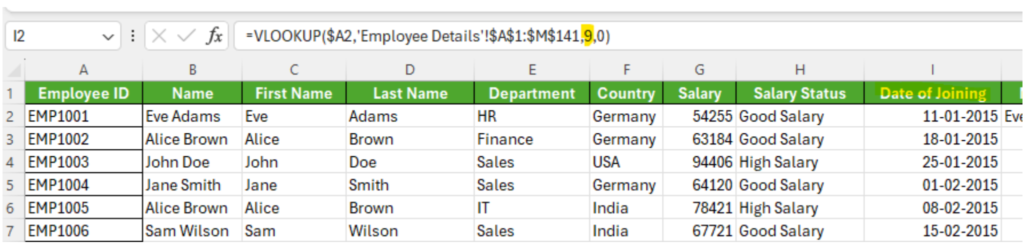
G2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,7,0)

	A	B	C	D	E	F	G
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184
4	EMP1003	John Doe	John	Doe	Sales	USA	94406
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120
6	EMP1005	Alice Brown	Alice	Brown	IT	India	78421
7	EMP1006	Sam Wilson	Sam	Wilson	Sales	India	67721



H2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,8,0)

	A	B	C	D	E	F	G	H
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary
4	EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary
6	EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary

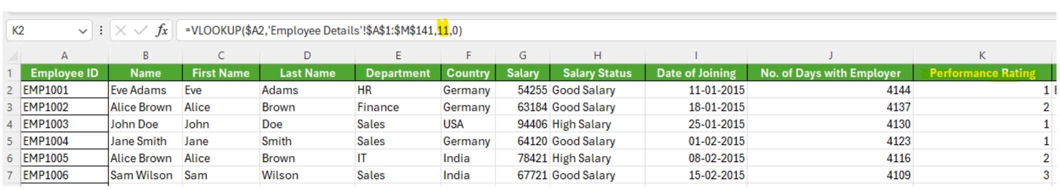


I2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,9,0)

	A	B	C	D	E	F	G	H	I
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary	11-01-2015
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary	18-01-2015
4	EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary	25-01-2015
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary	01-02-2015
6	EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary	08-02-2015
7	EMP1006	Sam Wilson	Sam	Wilson	Sales	India	67721	Good Salary	15-02-2015

J2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,10,0)

	A	B	C	D	E	F	G	H	I	J
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining	No. of Days with Employer
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary	11-01-2015	4144
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary	18-01-2015	4137
4	EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary	25-01-2015	4130
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary	01-02-2015	4123
6	EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary	08-02-2015	4116
7	EMP1006	Sam Wilson	Sam	Wilson	Sales	India	67721	Good Salary	15-02-2015	4109
8	EMP1007	Chris Lee	Chris	Lee	Marketing	India	35364	Low Salary	22-02-2015	4102



K2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,11,0)

	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining	No. of Days with Employer	Performance Rating
2	EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary	11-01-2015	4144	1
3	EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary	18-01-2015	4137	2
4	EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary	25-01-2015	4130	1
5	EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary	01-02-2015	4123	1
6	EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary	08-02-2015	4116	2
7	EMP1006	Sam Wilson	Sam	Wilson	Sales	India	67721	Good Salary	15-02-2015	4109	3

L2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,12,0)

Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining	No. of Days with Employer	Performance Rating	Gender
EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary	11-01-2015	4144	1	Other
EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary	18-01-2015	4137	2	Not Specifie
EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary	25-01-2015	4130	1	Male
EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary	01-02-2015	4123	1	Male
EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary	08-02-2015	4116	2	Female

Result : Complete Duplicate of Employee Details Table

M2 =VLOOKUP($A2,'Employee Details'!$A$1:$M$141,13,0)

Employee ID	Name	First Name	Last Name	Department	Country	Salary	Salary Status	Date of Joining	No. of Days with Employer	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve	Adams	HR	Germany	54255	Good Salary	11-01-2015	4144	1	Other	user106@example.com
EMP1002	Alice Brown	Alice	Brown	Finance	Germany	63184	Good Salary	18-01-2015	4137	2	Not Specif	user176@example.com
EMP1003	John Doe	John	Doe	Sales	USA	94406	High Salary	25-01-2015	4130	1	Male	user168@example.com
EMP1004	Jane Smith	Jane	Smith	Sales	Germany	64120	Good Salary	01-02-2015	4123	1	Male	user49@example.com
EMP1005	Alice Brown	Alice	Brown	IT	India	78421	High Salary	08-02-2015	4116	2	Female	user24@example.com
EMP1006	Sam Wilson	Sam	Wilson	Sales	India	67721	Good Salary	15-02-2015	4109	3	Male	user168@example.com

Day 14 – (Part 2)

Use: INDEX(), MATCH()

Task: Use INDEX and MATCH to fill missing values in one column by fetching the correct data from another related table.

I have used the match and index functions in previous step to correct the data

WEEK 3 – Visualization & Dashboard Building Focus: Pivot Tables, Charts, and Dashboard creation

Day 15 – Pivot Tables (Part 1)

● Connect multiple tables; if possible, do data modeling.

● Create simple PivotTables. Task: Show total sales by region or average marks by category.

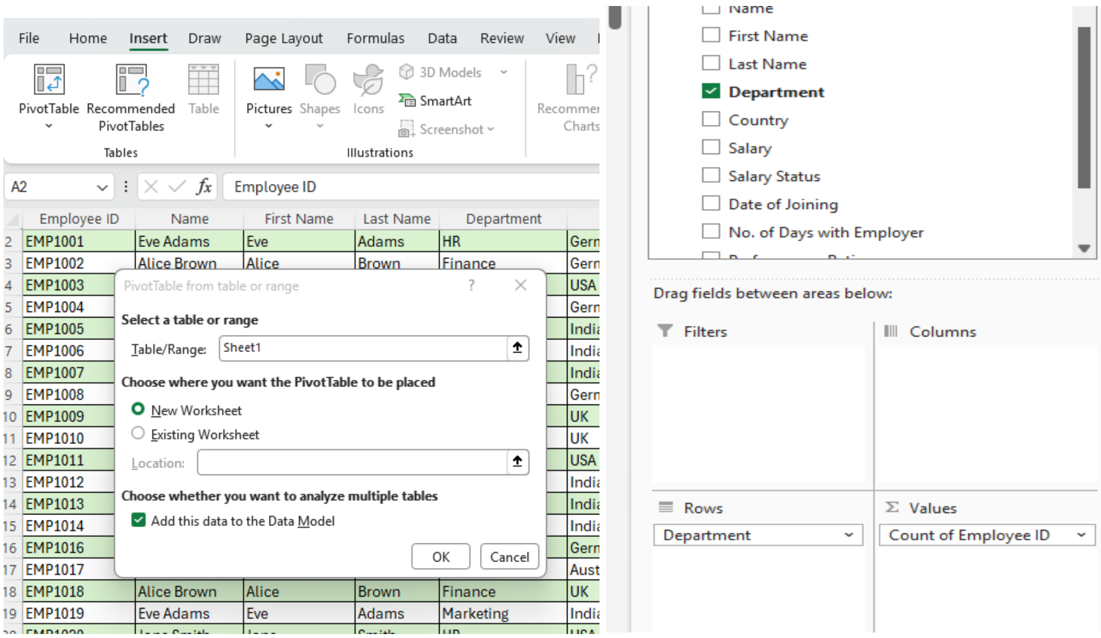
Pivot tables:

Department-wise Employee Count

In my employee details dataset, I have created a pivot table for department-wise employee count by following the below steps:

- Select the entire table
- Goto -> Insert -> Pivot table ->
- In the pivot table dialogue box -> Choose the sheet you have the dataset, select new or existing worksheet -> Check in Add this date to data model -> OK

- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **rows as Department and Values as Employee id.**



Department-wise Employee Count	
Row Labels	**Count of Employee ID**
Finance	21
HR	22
IT	22
Marketing	50
Sales	25
Grand Total	**140**

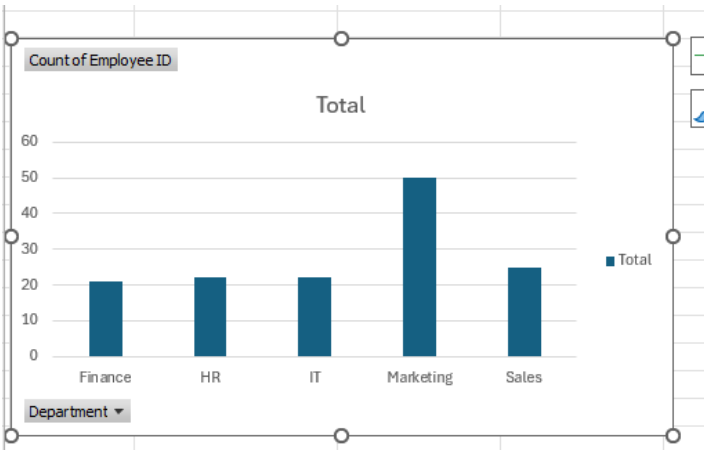
Day 16 – Pivot Tables (Part 2)

● **Create simple PivotTables.**

● **Write a short description of 4 types of analysis (descriptive, diagnosis, predictive, and prescriptive analysis) from the insights arrived from the pivot table**

● **Do forecast analysis Group data by date, category, etc. Task: Create one PivotChart linked to your PivotTable.**

- For the above created department wise pivot table, I have created a **Clustered column chart** as below,
- Select any cell in pivot table
- Goto Insert -> In charts tab -> Choose the column chart



Grouping – By Department

Insight - Shows workforce allocation across departments.

Analytics from Insights

Descriptive Analysis

"What happened?"

- Marketing and Sales have the highest employee count.

Diagnostic Analysis

"Why did it happen?"

- Business expansion may require more sales and marketing staff.

Predictive Analysis

"What may happen next?"

- If hiring continues similarly, other operational departments may become understaffed.

Prescriptive Analysis

"What should be done?"

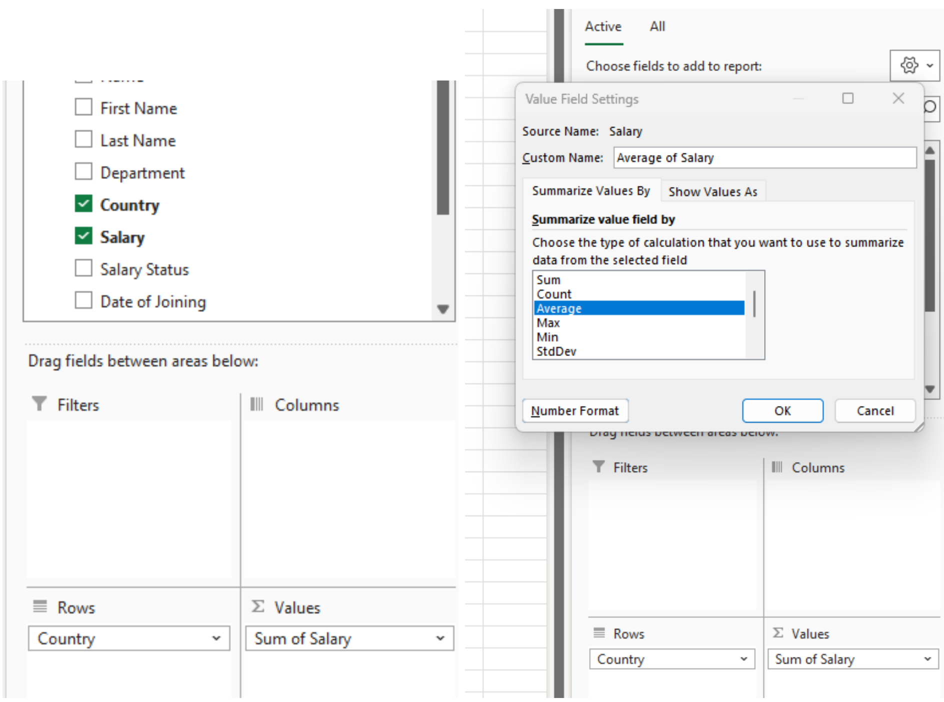
- Balance recruitment across departments.

Day 17 – Charts & Graphs (Part 1)

Practice: Bar, Column, Pie, Line charts. Task: Visualize key KPIs (such as total sales or average score).

Country-wise Average Salary Analysis

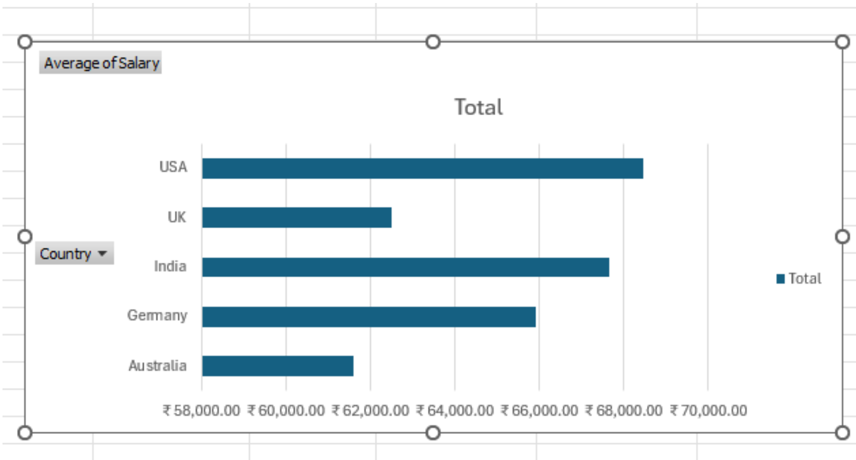
- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **Rows as Country and Values as Salary.**
- As we need the average salaries, I have selected the drop down of sum of salary and select -> field value settings -> choose average -> OK



Country-wise Average Salary Analysis

Row Labels	Average of Salary
Australia	₹ 61,609.84
Germany	₹ 65,897.32
India	₹ 67,654.17
UK	₹ 62,511.05
USA	₹ 68,461.57
Grand Total	**₹ 65,882.23**

- For the above created country wise pivot table, I have created a
- **Horizontal Bar chart** as below,
- Select any cell in pivot table
- Goto Insert -> In charts tab -> Choose the bar chart



Grouping - By Country

Insight - Compare compensation patterns globally.

Analytics from Insights

Descriptive

USA and India employees receive higher average salaries.

Diagnostic

Higher living costs and market standards affect salary structures.

Predictive

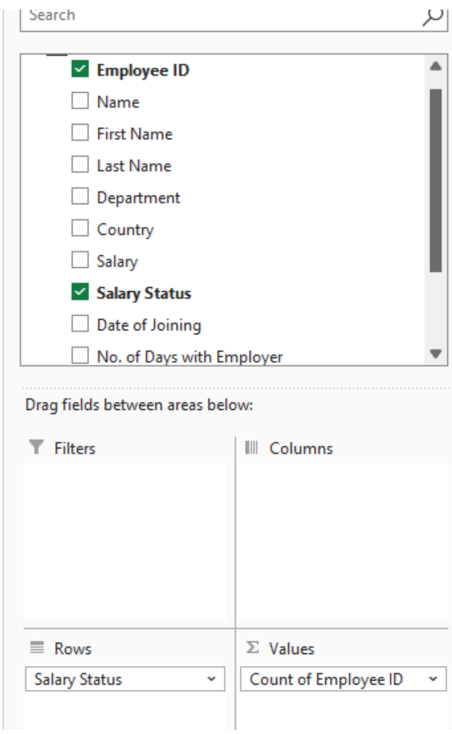
Salary budgets in developed countries may rise faster.

Prescriptive

Review compensation parity and optimize global salary strategy.

Salary Status Distribution

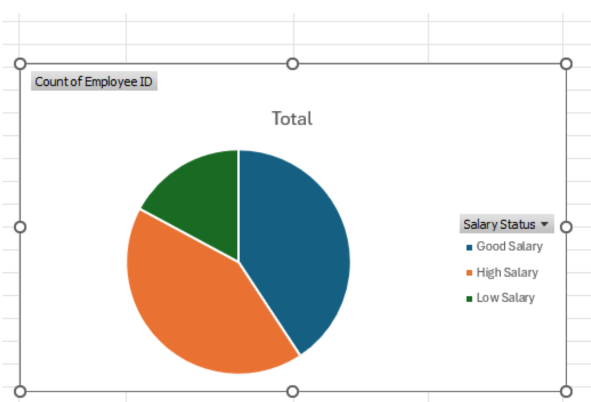
- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **Rows as Salary Status and Values as Employee id.**
- As we need the count of employee id, I have selected the drop down of count of employee id and select -> field value settings -> count -> OK



Salary Status Distribution

Row Labels	Count of Employee ID
Good Salary	57
High Salary	59
Low Salary	24
Grand Total	**140**

- For the above created salary wise pivot table, I have created a **Pie chart** as below,
- Select any cell in pivot table
- Goto Insert -> In charts tab -> Choose the pie chart



Grouping – Salary Status

Insight - Shows salary category distribution.

Analytics

Descriptive

Most employees fall under "High Salary."

Diagnostic

Salary bands are concentrated around mid-level roles.

Predictive

A rise in experienced hires may increase "Low Salary" count.

Prescriptive

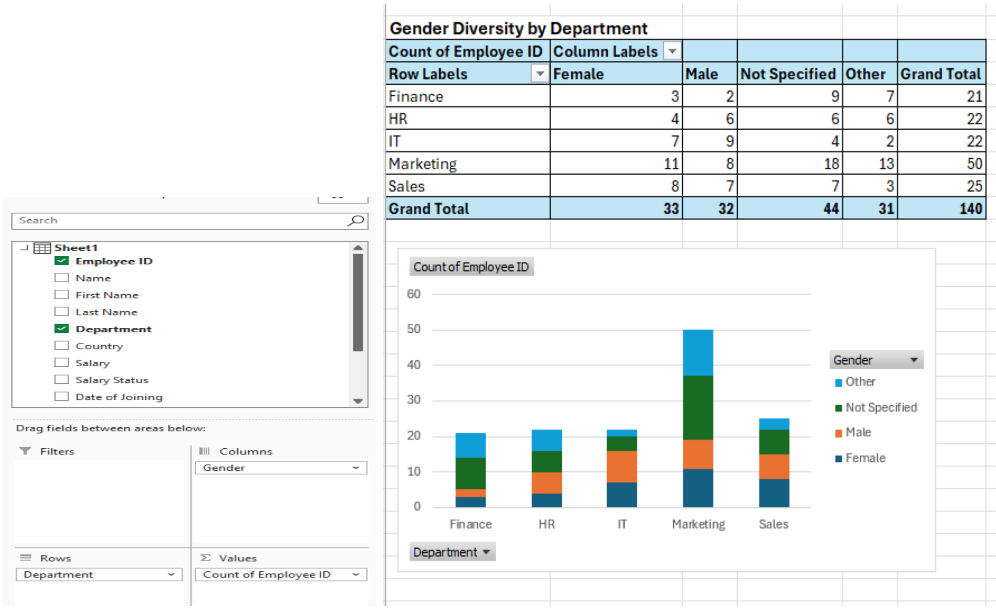
Introduce performance-linked salary planning.

Day 18 – Charts & Graphs (Part 2)

Create Combo charts, Sparklines, and use filters/slicers. Task: Add interactivity to your visuals.

Gender Diversity by Department

- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **Rows as Department, Columns as Gener and Values as Employee Id**



Gender Diversity by Department

Count of Employee ID	Column Labels				
Row Labels	Female	Male	Not Specified	Other	Grand Total
Finance	3	2	9	7	21
HR	4	6	6	6	22
IT	7	9	4	2	22
Marketing	11	8	18	13	50
Sales	8	7	7	3	25
Grand Total	33	32	44	31	140

Grouping – Department and Gender

Insight - Measures workforce diversity.

Analytics

Descriptive

Some departments show male/female imbalance.

Diagnostic

Historical hiring trends or role preferences may influence diversity.

Predictive

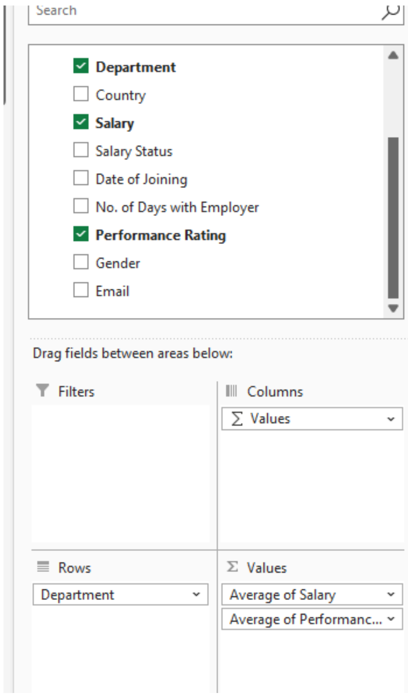
Without policy changes, imbalance may continue.

Prescriptive

Implement diversity hiring initiatives.

Department-wise Performance vs Avg Salary

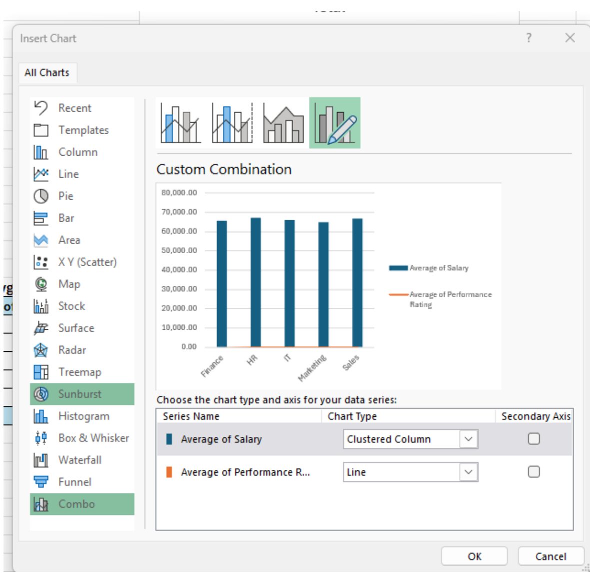
- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **Rows as Department, Values as Average of Salary and Performance Rating**

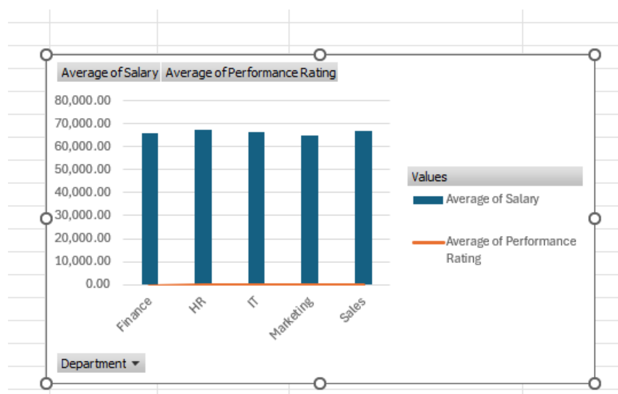


Department-wise Performance vs Avg Salary

Row Labels	Average of Salary	Average of Performance Rating
Finance	65,617.76	2.67
HR	67,234.18	2.73
IT	66,184.41	2.86
Marketing	64,847.26	3.10
Sales	66,718.68	2.84
Grand Total	**65,882.23**	**2.89**

- For the above department wise performance vs average salary pivot table, I have created a **combo chart** as below,
- Select any cell in pivot table
- Goto Insert -> In charts tab -> Choose the combo chart
- Select average of salary – column chart
- Select average of performance Rating – Line Chart





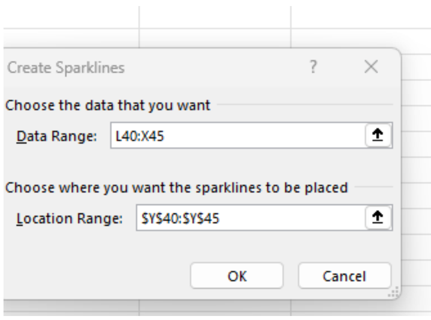
Insights:

Helps identify:

- High-paying but low-performing departments
- High-performing departments deserving salary revisions

Monthly Hiring Trend by Department

- Choose the Department and Hiring month data
- Select the above data range -> Goto Insert tab -> Sparklines -> Choose Column
- Select the sparkline range you need -. OK



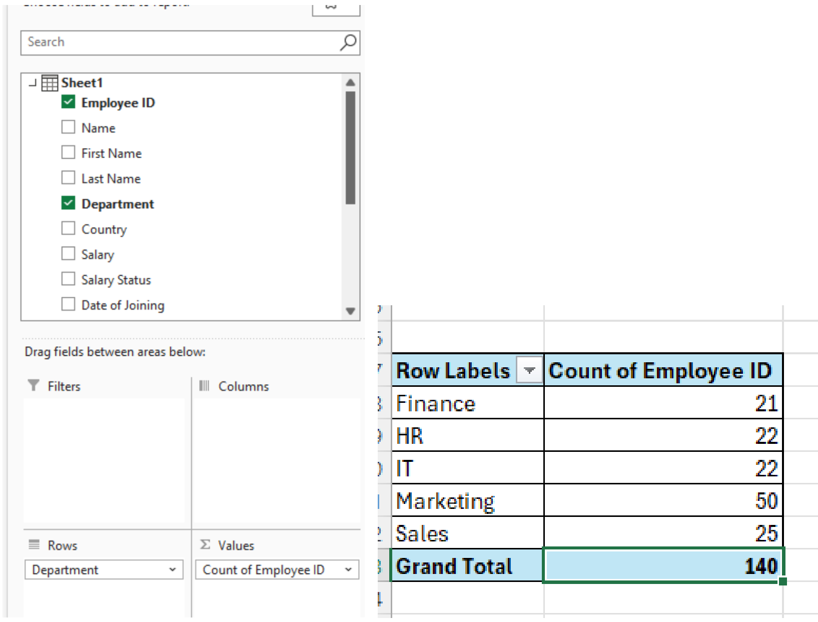
Department	April	August	December	February	January	July	June	March	May	November	October	September	Sparkline
Finance	2	2	1	2	2		3	1	4	2	2		
HR	1	3	2		2	3		3	2	2	2	2	
IT		1	1	2	1	2	1	5	1	2	3	3	
Marketing	4	3	3	4	5	5	6	3	4	4	4	5	
Sales	4	2	1	3	2	4	1	1	3		3	1	

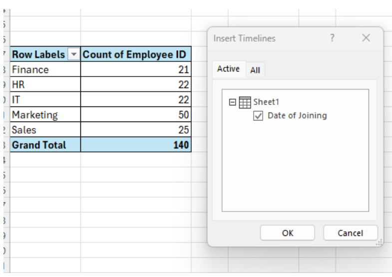
Employee and Department using timeline filter

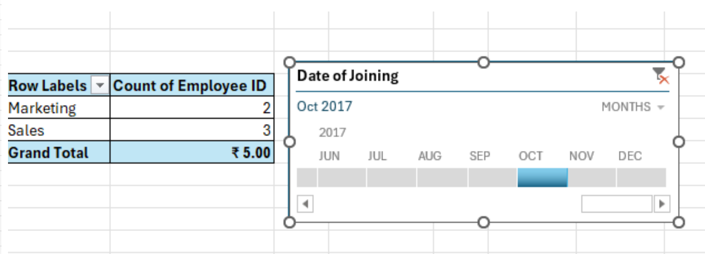
- In the pivot table field select and drag the columns you need for the pivot table for your analysis. Here, I have selected **Rows as Department, Values as Employee id**
- Select any cell in pivot table -> Inset -> Timeline filter
- In filter choose -> Date of joining

Benefits

- Interactive date filtering
- Analyze monthly/quarterly hiring trends
- Useful for forecasting dashboards





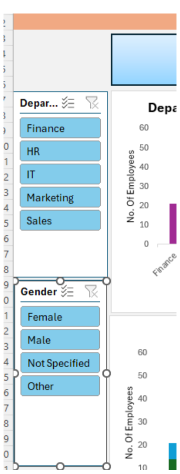


Slicers:

I have used the slicer once the dashboard created for the headers Department and Gender.

Select any chart

Goto -> Insert -> Slicer -> Choose the header we need slicer



Day 19 – Advanced Visualization & Storytelling

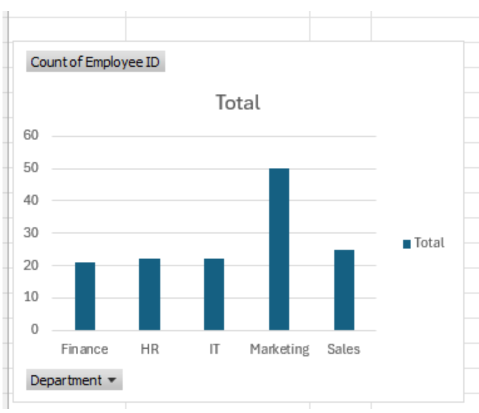
Add titles, legends, color themes, and storytelling flow. Task: Arrange visuals neatly and make it presentation-ready

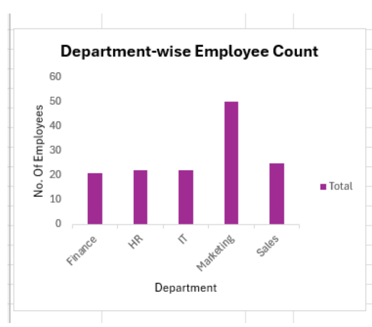
In order to make the above charts created, now I am going to apply some visualization effects to make them look good and easily analyzable

For example in the first chart - **Department-wise Employee Count**

- Select the chart you want to edit
- Select the buttons created, -> right click -> Hide All Field Buttons on chart
- Give Chart Title Name -> change font colour/style as per our requirements
- In the chart elements dialogue box -> Select Gridlines, Legend, Axis Titles

Before Editing the Chart: **After Editing the Charts:**



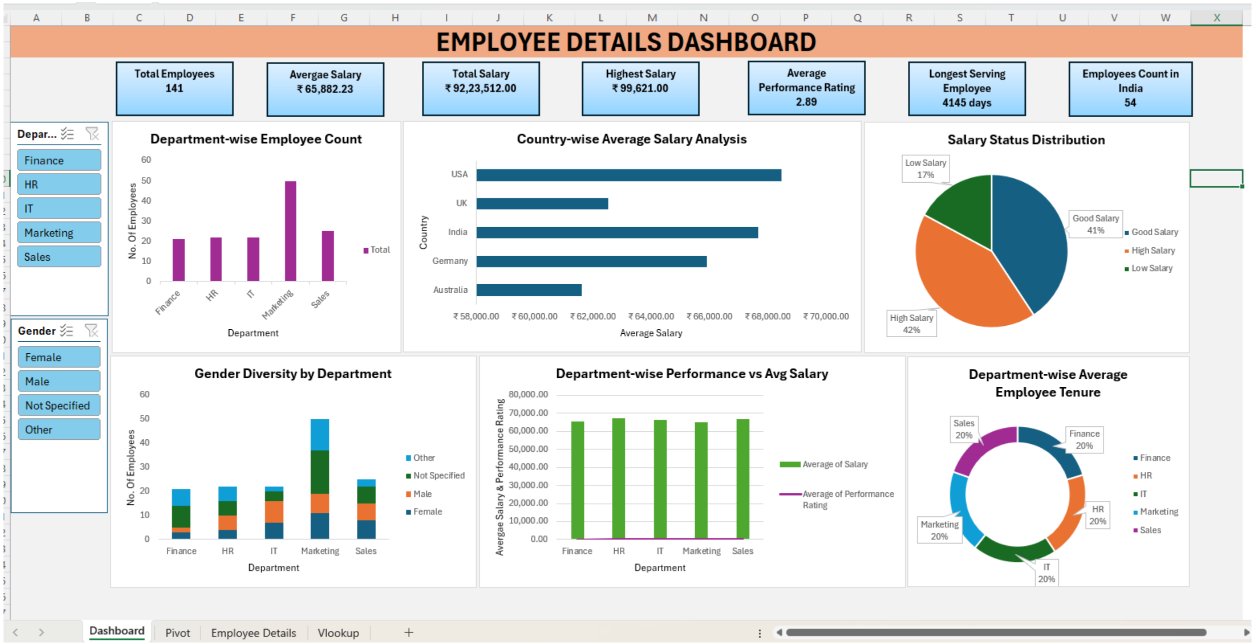


Day 20 – Dashboard Design & AI Integration

Use Quick Analysis for charts and summaries.

Try Excel Copilot or AI Analyst for formula suggestions and insights. Task: Make a one-page dashboard summarizing your insights.

I don't' have quick analysis and co-pilot in my excel version, therefore done the remaining analysis as per the above steps



Day 21 – Documentation & Submission

Employee Dataset Analysis Summary

1. Dataset Used

The dataset used for this analysis is an Employee Details dataset containing employee-related information such as:

Employee ID, Employee Name, Department, Country, Salary, Salary Status, Date of Joining, Number of Days with Employer, Performance Rating, Gender, Email ID

The dataset was analysed in Excel using Pivot Tables, Pivot Charts, Combo Charts, Sparklines, Slicers, Timeline Filters, KPI Metrics, and Forecast Analysis to derive HR and workforce insights.

2. Cleaning Steps Performed

The following data cleaning and preprocessing steps were performed before analysis:

Data Validation

- Checked for missing or blank values in important columns such as Employee ID, Salary, Department, and Performance Rating.

Duplicate Removal

- Removed duplicate employee records using Employee ID as a unique identifier.

Formatting Corrections

- Converted Date of Joining into proper date format.
- Standardized salary values into numeric format.
- Corrected inconsistent text entries in Department, Country, and Gender columns.

Data Categorization

- Grouped salary values into categories such as:
    - High Salary
    - Good Salary
    - Low Salary

Derived Columns Created

- Calculated No. of Days with Employer using joining date.
- Created Year and Month fields from Date of Joining for trend analysis.

Error Handling

- Removed invalid records and corrected formatting inconsistencies.

3. Formulas Used

The following Excel formulas and calculations were used in the analysis:

COUNTA, SUM, AVERAGE, MAX, MIN, COUNTIF, AVERAGEIFS

Additionally:

- Pivot Tables were used for aggregation and summarization.
- Forecast Sheet was used for predictive trend analysis.
- Sparklines were used to display mini trends.

4. Key Insights

Workforce Distribution

- Sales and Marketing departments had the highest employee count, indicating focus on business growth and customer acquisition.

Salary Analysis

- Average salary varied across countries and departments due to regional and organizational differences.
- Majority of employees belonged to the “Good Salary” category.

Performance Analysis

- Most employees had medium-to-high performance ratings, indicating stable workforce productivity.
- Certain departments showed stronger overall performance compared to others.

Diversity Analysis

- Gender distribution varied across departments, highlighting opportunities for diversity improvement in some teams.

Hiring Trend Analysis

- Employee hiring showed growth over time with noticeable recruitment spikes during specific periods.

Retention Analysis

- Departments with higher average tenure demonstrated better employee retention and stability.

Forecast Insights

- Forecast analysis predicted continued workforce growth and increasing payroll expenses in future periods.
- Hiring trends suggested future expansion in operational departments.

Dashboard Insights

- Interactive slicers and filters enabled dynamic analysis by:
    - Department
    - Country
    - Gender

- Salary Status
- Joining Year

Overall, the analysis provided actionable insights for workforce planning, salary optimization, performance management, diversity initiatives, and future HR strategy.

Final Dashboard